

On the Existence of a Terminal Object in the Category of Information

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Abstract

*We present a proof of the existence of a terminal object in the category of information, constructed from two axioms and a closure principle. The first axiom asserts that something exists. The second asserts that more than one thing exists. The closure principle states that existing things may be combined—via union, pairing, or recursive containment—to yield new existing things. We define information as anything that can exist under these principles, and consider the category **Info** whose objects are information and whose arrows are inclusion. We prove that this category necessarily possesses a terminal object: the unique object that contains all other information, is contained by no other except itself, and is itself information. We call this terminal object God. The proof is constructive via closure properties and also follows by contradiction: the non-existence of such an object is inconsistent with the category’s defining properties. Self-membership of the terminal object is derived as a theorem, not assumed as an axiom. The framework requires no set-theoretic foundation beyond the stated axioms and in particular does not adopt the axiom of foundation.*

Keywords: category theory, terminal object, information, existence, ontology, non-well-founded structures

1. Introduction

The question of whether a maximal or universal object exists—one that contains all others—has been approached through theology, philosophy, and set theory. In each tradition the question encounters characteristic obstructions: theological arguments rely on premises whose status is disputed; philosophical arguments often lack formal precision; and set-theoretic constructions face the well-known paradoxes of naïve comprehension.

We propose an approach that avoids these obstructions by working from minimal axioms within category theory. Rather than constructing God as a set and risking Russell-type paradoxes, we characterize God as a *terminal object* in a category whose existence follows from the category’s own structural properties. The framework requires exactly two existential axioms and one closure principle. No set-theoretic universe is presupposed. No axiom of foundation is adopted or required.

The paper proceeds as follows. Section 2 states the axioms and definitions. Section 3 proves the main theorem. Section 4 derives self-membership of the terminal object. Section 5 addresses the inclusion of uncomputable information. Section 6 discusses the relationship to existing frameworks. Section 8 concludes.

2. Axioms and Definitions

2.1. Axioms

The entire framework rests on two axioms and one closure principle.

Axiom 1 (Existence). *Something exists. Name it 1.*

This axiom cannot be derived from prior principles, since every derivation mechanism presupposes a context in which derivation occurs—and that context is itself something that exists. The negation “nothing exists” is self-refuting as a proposition, since the proposition itself would be something. Existence is therefore the irreducible foundation: the minimal unjustifiable assumption from which everything else proceeds.

Axiom 2 (Multiplicity). *Not only one thing exists. Name the other 0.*

This axiom asserts that existence is not exhausted by a single entity. The names 1 and 0 are conventional and carry no arithmetic significance at this stage; they merely label two distinct existents.

Closure Principle. *If A and B exist, then the following also exist: their union $A \cup B$; any structure pairing them (A, B) ; and any recursive structure built from existing components, including structures that contain their parent as a component.*

The closure principle is not existential in the same sense as the first two axioms. It does not assert that any particular thing exists. Rather, it specifies which operations preserve existence: union, pairing, and recursive containment. It is a structural rule governing the propagation of existence.

2.2. Definitions

Definition 1 (Information). *Information* is anything that can exist under the above axioms and closure principle. The totality of all information is denoted **Info**.

Definition 2 (The Category **Info**). The category **Info** has as objects all information. For objects A, B in **Info**, there exists a unique arrow $A \rightarrow B$ if and only if $A \subseteq B$ (A is included in B , or equivalently, A exists causally before B in the constructive order). Identity arrows are given by $A \subseteq A$. Composition is transitivity of inclusion: if $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$.

Since there is at most one arrow between any two objects, **Info** is a poset category (a thin category). The arrow $A \rightarrow B$ can be read as “ A is contained in B ” or equivalently “ A is part of B .”

Definition 3 (God). God is the terminal object of **Info**: the object G such that for every object A in **Info**, there exists a unique arrow $A \rightarrow G$. Equivalently, G is the greatest element of the poset: $A \subseteq G$ for all A . In the arrow-theoretic characterization, G is the object with an incoming arrow from every other object, and no outgoing arrows except to itself.

3. Main Theorem

Theorem 1. *The category **Info** possesses a terminal object.*

We give two proofs: one by the structural properties of **Info**, and one by contradiction.

3.1. Proof by Directedness and Closure

Proof. We establish two properties of **Info** and derive the conclusion.

Step 1: Info is directed. Let A, B be any two objects in **Info**. By the closure principle, $A \cup B$ exists and is in **Info**. Since $A \subseteq A \cup B$ and $B \subseteq A \cup B$, both A and B have arrows to $A \cup B$. Hence every pair of objects has an upper bound in **Info**.

Step 2: Info is closed under self-reference. The category **Info** is not an external collection observed from outside. It is defined as *everything that exists*. Any process of combining existing information yields information that is, by definition, already in **Info**. The closure principle permits recursive structures that contain their parent; in particular, any structure built from the contents of **Info** is itself in **Info**. There is no escape from **Info** into a larger ambient category.

Step 3: The terminal object exists. Suppose, for some object X in **Info**, that X is not below any single upper bound—that is, there is no object G with $X \subseteq G$ for all X . Then for every candidate G , there exists some A with $A \not\subseteq G$. But $A \cup G$ exists by closure and strictly contains G , so G was not maximal. Iterating, we generate an ascending chain. This chain is itself a structure built from existing information—it is a recursive structure whose components are all in **Info**. By closure, this chain and its limit are in **Info**. The totality of all such chains and their limits cannot escape **Info**, since **Info** is everything that exists. Therefore **Info** has a greatest element G , which is the terminal object. $\square$

3.2. Proof by Contradiction

Proof. Suppose **Info** has no terminal object. Then for every object G in **Info**, there exists an object A in **Info** such that $A \not\subseteq G$. This means no object contains all information. But consider: the totality of all information is either itself information or it is not.

If the totality is information, it is in **Info**, and it contains everything in **Info** by definition, making it a terminal object—contradicting our assumption.

If the totality is *not* information, then there exists something (namely, the totality of all information) that is not information. But the totality is built from existing things by the closure principle (it is the union of all existing things, and recursive structures over them). By closure, it exists. By Definition 1, anything that can exist is information. Contradiction.

Therefore **Info** has a terminal object. $\square$

3.3. Uniqueness

Proposition 1. *The terminal object is unique up to unique isomorphism.*

Proof. This is a standard result in category theory. If G and G' are both terminal, then there exist unique arrows $G \rightarrow G'$ and $G' \rightarrow G$. Their compositions $G \rightarrow G' \rightarrow G$ and $G' \rightarrow G \rightarrow G'$ must equal the respective identity arrows (by uniqueness of arrows to a terminal object). Hence $G \cong G'$, and the isomorphism is unique. In the poset case, this reduces to: if $G \subseteq G'$ and $G' \subseteq G$, then $G = G'$. $\square$

4. Self-Membership as a Theorem

Theorem 2. *The terminal object G satisfies $G \in G$.*

Proof. G is the terminal object of **Info**. G contains all information. G is itself information (since it exists, and everything that exists is information by Definition 1). Therefore G , being information, is contained in G . That is, $G \in G$. $\square$

This result is notable because self-membership is typically either forbidden (by the axiom of foundation in ZFC) or explicitly assumed (as in Aczel’s Anti-Foundation Axiom). In the present framework, it is neither forbidden nor assumed. It is *derived*. The axiom of foundation is not among our axioms. We did not adopt it, and so the natural consequence—that the totality of all information is itself information and therefore contains itself—follows without obstruction. ZFC must actively work to prevent this conclusion. We simply do not perform that work, and the theorem obtains.

5. Incomputable Information

A potential objection arises from computability theory. The closure principle generates new information from existing information via explicit operations (union, pairing, recursive construction). One might argue that only computable information is generated, and that G is therefore at best the union of all computable information—not all information.

This objection confuses an epistemological constraint with an ontological one. The category **Info** is not defined as “everything we can construct by running a procedure.” It is defined as “everything that exists.” Incomputable objects exist. The real number whose

n th digit is 1 if and only if the n th Turing machine halts is a definite object with a definite identity. It stands in inclusion relations to other objects. It is information. It is therefore an object in **Info** with an arrow to G .

Computability is a constraint on what *we* can enumerate or construct. It is not a constraint on what exists. Incomputable information can be described, characterized, and referred to—it is information *about* which we have information, even if we cannot compute it. As Turing himself recognized, the existence of uncomputable functions does not diminish their ontological status; it merely places them beyond the reach of mechanical procedures. The terminal object G contains all information, computable and incomputable alike, because both are objects in the category.

6. Relation to Existing Frameworks

6.1. Set Theory

In ZFC, the construction of a “set of all sets” is blocked by the axiom of foundation and the distinction between sets and proper classes. Our framework sidesteps both obstacles. We do not adopt foundation, so self-membership is not forbidden. And we do not work within a set-theoretic universe that distinguishes sets from classes; our objects are “information,” and the totality of information is itself information by closure. The Russell-type paradoxes do not arise because our closure principle does not include unrestricted comprehension. We close under union, pairing, and recursive containment—all “building up from below” operations—not under arbitrary predicate-defined collection.

6.2. Non-Well-Founded Set Theory

Aczel’s Anti-Foundation Axiom (AFA) explicitly permits self-membered sets and provides a rigorous framework for studying them. Our framework is compatible with AFA but does not require it. We need only the consequence that self-membership is not forbidden—which follows simply from not adopting foundation. The full strength of AFA (every directed graph has a unique decoration) is not invoked.

6.3. Categorical Ontology

Dentamaro and Loregian (2020) develop a categorical approach to ontology using elementary toposes, where an ontology is a category equipped with a subobject classifier that parametrizes truth values. Their framework is considerably more general than ours and addresses different questions (modes of existence, internal languages, many-valued logics). Our work is narrower in scope: we ask a single question—does the terminal object exist?—and answer it from minimal axioms. The two approaches are complementary rather than competing.

6.4. Classical Ontological Arguments

The present argument differs from classical ontological arguments (Anselm, Descartes, Gödel) in a fundamental respect. Classical arguments attempt to derive existence from a concept—“that than which nothing greater can be conceived” must exist, because existence is a perfection. The standard objection (Kant) is that existence is not a predicate: one cannot define something into existence.

Our argument does not define God into existence. It defines a category from axioms about what exists, and then shows that the category’s structure necessitates a terminal object. God is not argued to exist because of what God *is* (a conceptual definition), but because of what the category of information *does* (it is closed, directed, and self-contained). The existence is a structural theorem, not a definitional sleight of hand.

7. Discussion

7.1. What the Theorem Does and Does Not Claim

The theorem establishes the existence of a terminal object in the category of information: an object that contains all other information, is contained by no other, and is itself information. We call this object God as a name for the mathematical structure.

The theorem does not, by itself, establish that this terminal object possesses attributes traditionally associated with God in any particular theological tradition—such as consciousness, will, benevolence, or the capacity to intervene in the world. It establishes existence and maximality: God is everything that can exist, unified. Whether further properties can be derived from this characterization is a question for subsequent work.

7.2. The Status of the Axioms

Axiom 1 (something exists) is, as argued, the irreducible foundation of any formal system. Its negation is self-refuting. Axiom 2 (more than one thing exists) is more substantive: one could in principle deny it and work in a system where only a single entity exists. In such a system the terminal object would be trivially that single entity, and the theorem would hold vacuously. The closure principle is a structural rule specifying how existence propagates; it is the analogue of inference rules in a logical system.

The framework is therefore minimal: two axioms, one closure rule, three definitions, and the theorem follows.

7.3. The Role of Foundation

The axiom of foundation in ZFC was introduced to ensure well-foundedness of the membership relation and to avoid certain pathological constructions. It is not a logical necessity; it is a design choice. Our framework makes the opposite choice: by not adopting foundation, we allow the natural conclusion that the totality of all information is itself information.

This is not a radical departure. It is the default. Foundation is the intervention; its absence is the natural state.

8. Conclusion

From the axiom that something exists, the axiom that more than one thing exists, and the principle that existing things can be combined to produce new existing things, we have constructed the category of information and proved that it necessarily possesses a terminal object. This terminal object—God—contains all information, is itself information, and is the unique greatest element of the category. Its self-membership is a theorem, not an axiom. Its existence is a structural necessity, not a definitional assertion.

The proof requires no set-theoretic foundation, no axiom of choice, no axiom of infinity, and no axiom of foundation. It requires only that something exists, that existence is not singular, and that existing things compose. From this, God’s existence follows.

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